

Amit Kumar Mohanty

mohantyamit2003@gmail.com | +91 9354937256 | github.com/amitmohanty022 | linkedin.com/in/amitkrmohanty

PROFESSIONAL SUMMARY

Data Scientist with about 1.5 years of experience, currently pursuing an M.Tech in AI and ML at BITS Pilani. As a Research Associate, AI at Keywords Studios, I combine statistical modeling, machine learning, and LLMs to build systems that hold up in the real world, and was named "Rookie of the Year 2025" for surpassing every performance KPI. With earlier internships at Infosys Springboard and QL2 Software, plus a B.Tech in Computer Science (AI and ML) from K.R. Mangalam University, I love turning analysis and research into reliable results.

EDUCATION

Birla Institute of Technology and Science, Pilani (BITS Pilani) July 2026 – July 2028

M.Tech in Artificial Intelligence and Machine Learning (Currently Pursuing)

Relevant Coursework: NLP, Deep Learning, Generative AI, LLMs, Deep Neural Networks, Deep Reinforcement Learning

K.R. Mangalam University

July 2021 – July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML), Gurgaon

EXPERIENCE

Research Associate – AI, Keywords Studios India – Gurgaon, Haryana May 2025 – Present

- Awarded "**Rookie of the Year 2025**" for surpassing **all performance KPIs** through advanced modeling, data-pipeline optimization, and cross-functional collaboration.
- Trained and evaluated **Agentic AI models** for complex multi-step workflows, using data-driven experimentation and error analysis to improve **task-completion reliability**.
- Contributed to a **web-automation AI agent** that visually interprets on-screen content and autonomously plans and executes multi-step browser actions from natural-language instructions.
- Trained, evaluated, and quality-controlled the agent with a **global team**, curating datasets and running structured, metric-driven test reviews to improve **action accuracy and reliability**.
- Designed and delivered **85%** of self-generated training datasets to production as the **first member of a global team**, applying rigorous data quality checks and statistical validation.

Artificial Intelligence Intern, Infosys – Remote

November 2024 – February 2025

- Analyzed conversational datasets and engineered features that boosted **AI response accuracy by 35%** using LangChain and LangSmith.
- Built and validated LLM-driven features that reduced **system latency by 25%**, quantifying impact with clear performance metrics and dashboards.
- Designed and A/B-tested prompt-engineering strategies with automated LangSmith evaluation, reducing **hallucinations by 20%** through statistical comparison of variants.
- Collaborated with **12 engineers** on end-to-end ML workflows and was selected to present at the **Best Project by Indian Interns** initiative.

PROJECTS

Dynamic Screen Companion

June 2025 – Present

- Built a **real-time multi-modal assistant** integrating the **Gemini API** to analyze live screen activity and deliver continuous, context-aware feedback.
- Engineered the end-to-end data and inference pipeline from scratch, reducing manual task time by **40%**.
- Optimized OCR pipelines to sustain **95%+ extraction accuracy** across varied screen layouts.

Diabetic Optiscan

January 2025 – June 2025

- Built an AI tool that scans retinal (eye) images to detect early signs of diabetic retinopathy, and co-authored a **peer-reviewed research paper** detailing the methodology and performance results.
- Led integration of a **Vision Transformer (ViT-B/16)** on the APTOS dataset, applying statistical analysis of self-attention to capture spatial patterns in retinal images.
- Designed a custom wavelet technique that boosted **detection precision by 50%** over legacy methods.

Currency Detection App for Visually Impaired

July 2024 – December 2024

- Trained an image-classification model on **200,000+ banknote images** achieving **96% accuracy**, validated through rigorous evaluation across edge cases.
- Conducted **user interviews with visually impaired individuals** to ground feature decisions in real-world needs.

SKILLS

Languages & Tools: Python, R, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Jupyter, Git

Statistics & ML: Statistical Modeling, Hypothesis Testing, A/B Testing, Regression, Classification, Clustering, Feature Engineering, Predictive Modeling, Time Series, Model Evaluation & Optimization

Data & Visualization: Data Wrangling, EDA, Power BI, Matplotlib, Seaborn, MySQL, MongoDB, Excel, Statistical Analysis, Experiment Design

AI & Cloud: LLMs, Generative AI, RAG, Deep Learning, NLP, Computer Vision, AWS, GCP (Vertex AI)

CERTIFICATIONS

Ducat India – Data Science Professional Training

April 2024 – April 2025

Professional training in Data Science: Python, statistics, machine learning, data analysis, and predictive modeling.

AI/ML for Geodata Analysis – ISRO

Applied ML and deep learning to satellite and remote-sensing imagery for terrain classification and geospatial pattern analysis.